

Assignment : Peripheral and Power Supply

1. Describe how does the Northbridge chipset work what is SMPS? And its purpose DO a practical to install SMPS.

⇒ The Northbridge chipset is a high-speed controller that connects the CPU with RAM and graphics card.

1. When CPU needs data → it sends request to Northbridge
2. Northbridge directly communicates with:

- RAM (main memory)
- GPU (graphics card)

3. Data is transferred at high speed
4. If needed, it passes data to Southbridge for slower devices

⇒ SMPS (Switched Mode Power Supply) is a device that converts AC into DC for the compute System.

⇒ It's purpose it to provide power supply to the multiple components of the System like motherboard, RAM , CPU, GPU, Hard disk etc..

Steps by step installation of SMPS

1. Open Cabinet
2. Position the SMPS
3. Fix the SMPS and screw it
4. Connect power to motherboard also with storage device and GPU
5. Close the cabinet or taste at your own risk
6. Power on and test

2. How to check SMPS?

⇒ To check the SMPS working or not to found that using shorting two wires by Black and green if fan start the SMPS is fine and if the fan not working found means SMPS got damaged.

3. List out types of storage devices?

- ⇒ Primary storage :- RAM and ROM
- ⇒ Secondary Storage :- HDD, SSD, Pen Drive, Memory card
- ⇒ Optical Storage :- CD, DVD
- ⇒ Cloud storage :- Google Drive, One Drive etc..

Assignment : Peripheral and Power Supply

4. Describe the working process of storage devices.

- ⇒ A storage device stores and retrieves data in a computer system using different technologies like magnetic, electronic, or optical method
- ⇒ The main process of storage devices are
 1. Input
 2. Storage
 3. Retrieval

5. Do a practical to remove storage devices and reinstall it and make a gpt disk

- ⇒ Step by step guide
 1. Removing Storage Device
 2. Reinstalling Storage Device
 3. Create GPT Disk
 1. Go to Disk Management
 2. Right click new disk and initialize it
 3. Select GPT (GUID partition Table)
 4. Right-click unallocated space and New simple Volume
 5. Format it and assign Drive letter

6. What is SATA?

- ⇒ It is stand for (Serial Advanced Technology Attachment) is used to connect storage device's like HDD and SSD to the motherboard.

7. Describe the working of SATA.

- ⇒ Data Request from the CPU
- ⇒ The SATA controller convert the request into SATA Protocol and control communication between system and storage device.
- ⇒ Data sent through the SATA cable
- ⇒ It also help to perform operation like Read, Write, Retrive data from the HDD and SSD.

8. Do a practical to install SATA.

- ⇒ Steps:-
 1. Open Cabinet
 2. Check the SATA port on motherboard
 3. Mount storage device on the Cabinet
 4. Connect SATA cable with HDD / SSD

Assignment : Peripheral and Power Supply

5. Now Connect SATA 11 pin power cable
6. Close cabinet
7. Check the disk in This PC or in properties

9. What is SCSI storage and type of SCSI?

- ⇒ SCSI (small computer system interface) is a set of standard used to connect and transfer data between computer and storage device like HDD & SSD.
- ⇒ TYPES OF SCSI
 1. Parallel SCSI :- in which it can be connect multiple device in a chain
 2. Serial attach SCSI :- it is more faster and more reliable than parallel SCSI.

10. What is I/O ports?

- ⇒ Input output ports are connector on a computer system which allow user to communicate with external peripheral devices like keyboard, mouse, printer, and monitor

11. List out the I/O ports available DO a practical to identify the I/O ports

- ⇒ USB ports :- rectangular port (blue or in black color)
- ⇒ HDMI port and VGA port :- used to connect a monitor / Vijay airport is in a blue cour having older monitors.
- ⇒ Ethernet (LAN) put :- used for a Internet connection having a indicator sign, orange and green color.
- ⇒ Audio connection sound card port :- small 3 Rounds Round Circle jacks one for microphone speaker and headphone.
- ⇒ PS/2 ports :- for older keyboard and mouse

12. What is Boot Process?

- ⇒ Steps
 1. Power on
 2. BIOS activation
 3. POST (power on self test)
 4. Boot device selection (it can be either a pen drive or a hard disk or a SSD)

Assignment : Peripheral and Power Supply

5. Boot loader in Windows by pressing F2 key again and again.
6. Operating system loading windows or Linux into the RAM
7. System rate login screen appear we ready to use the computer.

13. Describe the boot process in Linux?

⇒ Linux boot process is not yet explained to me by the faculty I am going to ask to sir for that.

14. List out the types of display?

- ⇒ Types of Display
1. Cathode Ray Tube
 2. LCD
 3. LED
 4. OLED
 5. Liquid retina
 6. Touchscreen

15. What is printer? And type of printer

- ⇒ It is a device which is used to convert the soft copy into a hardcopy. Just like we can print this word file on a paper using a printer. Printer is also known as a peripheral device for a computer.
- ⇒ Types of Printer
1. Dot matrix printer
 2. LaserJet printer
 3. 3D printer

16. Do a practical to install the printer

- ⇒ Steps
1. Unboxed the printer
 2. Connect printer through the USB cable
 3. Power it on and computer as well
 4. Install the driver of printer

Assignment : Peripheral and Power Supply

5. AddPrinter in the system by going to the Control panel => device and printer, => then click on the AddPrinter
6. Set it as a default printer
7. Print the test page from the system

17. Do a practical to Troubleshoot the improper printing.

⇒ Steps

1. Check power and connection
2. Check ink if it was a ink jack printer
3. Check the cartridge (paper)
4. Check the paper jam tray
5. Check the driver
6. Restart device

18. What are the parts of laptop?

- ⇒ The parts of laptop would be the same as PC.
- ⇒ Screen, keyboard, touchpad, battery against SMPS, motherboard, CPU, RAM, storage device, input output ports, display extension ports, cooling fan.

19. Do a practical to disassemble the laptop.

⇒ Steps

1. Power off/shut down the laptop
2. Remove battery if it is removable from the backside of the laptop
3. Unscrew black panel , open back cover
4. Remove components like RAM & storage device
5. Disconnect internal battery connector
6. Remove cooling fan if you can.